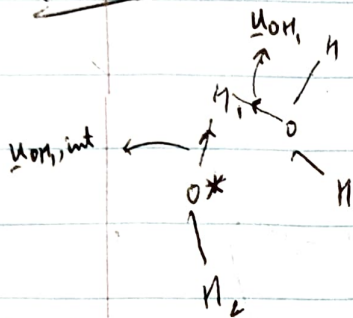


find the O-O distance b/w 2 O atoms.

The frame re-orientation: focuses mainly on the reorientation of the H-bond axis



Procedure

- do this x time steps
- Find the O-O vector ($\underline{u}_{OO}$) & then normalize it ($\underline{u}_{OO}^{norm}$)
 - Qualification for O-O part's H-bond: $|\underline{u}_{OO}| \leq \text{cut-off}_{OO}$
 - Find the O-H₁ vector ($\underline{u}_{OH_1}$) & then normalize it ($\underline{u}_{OH_1}^{norm}$)
 - Find the qualifying pairs using OH₁: $|\underline{u}_{OH_1}| \leq \text{cut-off}_{OH_1}$
 - Now, find the intramolecular $\underline{u}_{OH_1, int}$ & normalize it ($\underline{u}_{OH_1, int}^{norm}$)
 - Now, find the $\angle H_1 O^* O$ using the dot product.
 - Check for qualification for the angle part.
- again, x time frames.

for a given time frame

Now, combine all qualifications for H₁ case and find the O & H:

$$\text{qual_H1bond} = (\text{qual_OO}) \& (\text{qual_OH1}) \& (\text{qual_angle_OH1_OO})$$

So qual_H1bond: $\begin{matrix} N \times N \\ (\text{donor} \times \text{acceptor}) \\ N_d \times N_a \end{matrix}$ sized array

$\begin{matrix} N_d \times N_a \text{ acceptor} \\ (\text{donor} \times \text{donor}) \\ N_d \times N_d \end{matrix}$ & $\begin{matrix} N_d \times N_d \\ (\text{donor} \times \text{acceptor}) \end{matrix}$

& qual_H1bond: donor x acceptor

Similarly for H₂ bond

- now $\underline{u}_{OO}$ vectors to be tracked
- Extract all the O & H involved in the H-bonding process, sort them (this index, even skipping are "sorting" step is also fine)
 - Find the Hbonds i.e. @ time t, for a donor @ index = d2; what's the index of acceptor?
 - Find the $\underline{u}_{OO}^{norm}$: $O^* \leftarrow \text{donor Oxygen}$
 $O^a \leftarrow \text{acceptor Oxygen}$

Terminology: starting time t for a given block
 initial time t for finding correlation ($t=0$ case)
 final time t again, for correlation ($t=t$ case)

for the TCF now! Our aim is to get $\langle P_2 [u_{00}^{nom}(0), u_{00}^{nom}(t)] \rangle$

- ✓ for every block, determine the starting time for the block. Then, w/in every block, we'll have many ^{initial} starting times, based on the max^m # of initial times we can have.
- ✓ Extract all the values ~~for~~ of the H-bond $1/2$ @ the initial time (essentially - the H-bond indices @ $t = \text{initial time}$ for H_1 & H_2
- ✓ Corresponding to the H-bonds @ $t = \text{initial time}$, also find the u_{00}^{nom} (using the norm- u_{00} -H-bond $1/2$ array)
- ✓ $t = \text{final time} = \text{initial time} + K$: Now, for the final time also, track the H-bonds which are still intact!!
- ✓ at the final time, find the cosine of the angle formed by OO vector w/ the OO vector @ initial time (for the intact bonds only).
- ✓ find the P_2 & hence $\langle P_2 \rangle$ for all such angles.
- ✓ for as the ensemble average; we're to divide ~~over~~ P_2 by the # of H-bonds that existed @ initial time (Think of it like we're trying to find the probability that the bonds do ~~have~~ still exist, so we need to divide by the initial # of bonds)